Exercise 01: Master Theorem

Write a text file named '`master.txt`' that finds the runtime of each recurrence below using the master theorem. Show your work.

Recurrences

1. $T(n) = 4T(n/2) + n$
2. $T(n) = 4T(n/2) + n^2$
3. $T(n) = 4T(n/2) + n^3$
4. $T(n) = T(9n/10) + n$
5. $T(n) = 2T(n/4) + \sqrt{n}$
6. $T(n) = 7T(n/2) + n^2$